DGMS 2019 • SECTION G: SEISMOLOGY & EARTH INTERIOR

Question 62: Earth's Internal Structure, Seismic Boundaries & Core-Mantle Physics

Detailed Multi-Statement Verification, Seismological Discontinuities, and PREM Reference Data

OFFICIAL EXAMINATION QUESTION

Consider the following statements regarding the earth's interior:

1. Seismic waves reveal that the average depth of mantle is about 3100 km.

2. Based on the seismic wave studies, the concept of solid inner core was given by Inge Lehmann.

3. The sharp contrast from seismological wave velocities below the crust was noticed by Andrija Mohorovičić.

4. Transverse waves do not propagate in the outer core and hence the material here is in fluid stage.

Which of the statements given above are correct?

(a) 1 and 2 only

(b) **2, 3 and 4 only [CORRECT]**

(c) 1, 3 and 4 only

(d) 1, 2, 3 and 4

VERIFIED ANSWER KEY

Correct Option: (b) 2, 3 and 4 only — Statement 1 is factually incorrect because the core-mantle boundary (Gutenberg discontinuity) lies at a depth of ~**2890 km (2898 km**

PREM), not 3100 km. Statements 2, 3, and 4 are foundational, historically and physically accurate principles of global seismology.

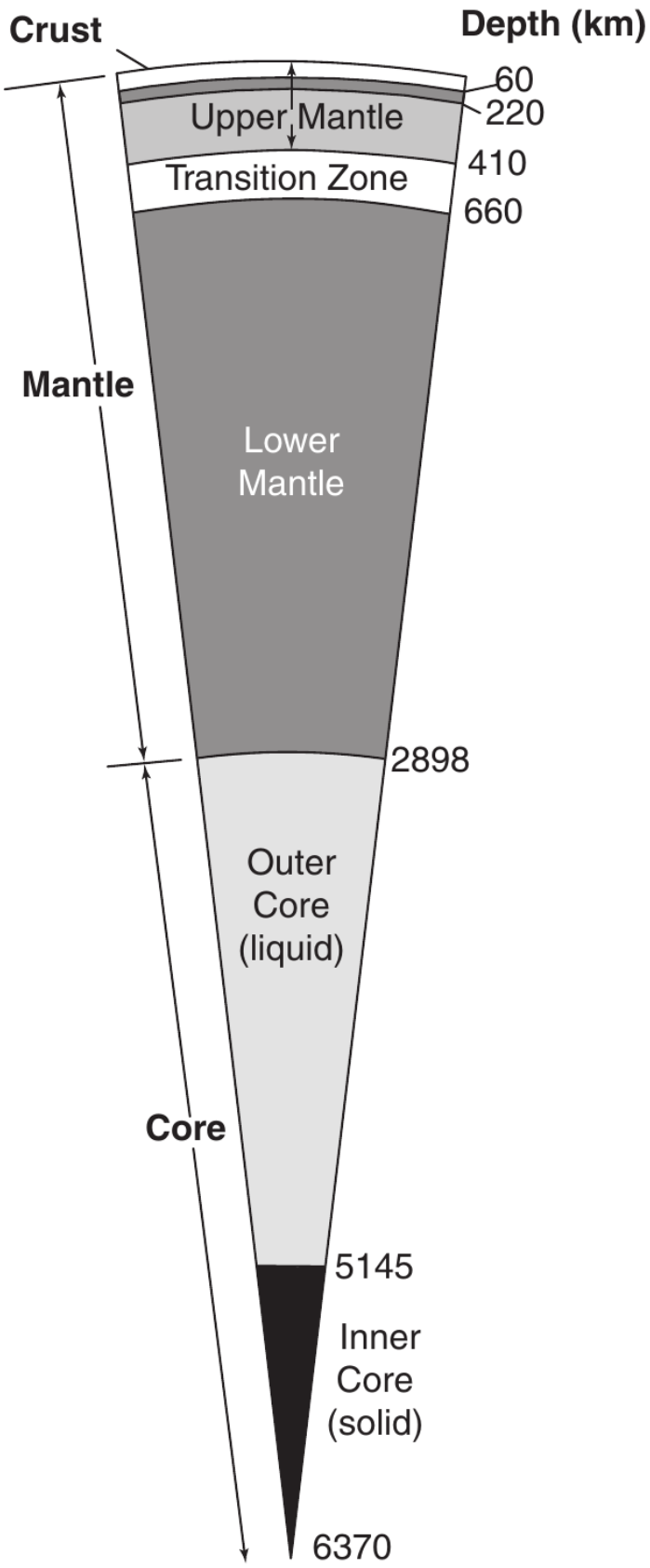


FIGURE 2 Major subdivisions of the Earth.

Figure 62.1 | Radial Subdivisions and Boundaries of the Solid Earth: Cross-section illustrating the exact seismological depth boundaries from PREM: Crust (0–35 km), Upper Mantle (to 410 km), Transition Zone (410–660 km), Lower Mantle (extending to the Core-Mantle Boundary at **2898 km**),

Liquid Outer Core (2898–5145 km), and Solid Inner Core (5145–6370 km). (Source: John D. Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd Ed., Fig. 1-2, p. 7).

1. Statement-by-Statement Diagnostic Breakdown

Statement 1: "Average depth of mantle is about 3100 km" — INCORRECT

The Earth's mantle extends from the base of the crust (Moho discontinuity, averaging ~30–35 km beneath continents and ~7 km beneath oceans) down to the **Core-Mantle Boundary (CMB)**, historically known as the *Gutenberg discontinuity*.

Extensive seismic refraction, reflection, and normal-mode studies—crystallized in the *Preliminary Reference Earth Model (PREM)* of Dziewonski & Anderson (1981)—demonstrate unequivocally that the CMB is situated at a depth of **2890 km (2898 km in PREM)**. A depth of 3100 km is factually incorrect by over 200 km.

Statement 2: "Solid inner core discovered by Inge Lehmann" — CORRECT

Prior to 1936, the Earth's core was believed to be an entirely molten iron sphere. In 1936, Danish pioneering seismologist **Inge Lehmann** published her landmark paper titled *"P'"* (P-prime). She analyzed faint P-wave arrivals recorded in the European and New Zealand shadow zone ($103^\circ < \Delta < 143^\circ$) from an earthquake near Buller, New Zealand (1929).

She proved that these arrivals were waves reflected ($PKiKP$) and refracted ($PKIKP$) by a distinct, higher-velocity inner core boundary situated at a depth of **~5150 km (5145 km)**. Harold Jeffreys and Keith Bullen subsequently confirmed that the high compressional velocity and shear wave transmission require the inner core to be solid.

Statement 3: "Sharp velocity contrast below crust noticed by Andrija Mohorovičić" — CORRECT

In 1909, Croatian seismologist and meteorologist **Andrija Mohorovičić** analyzed seismograms from the Pokuplje (Kulpa Valley) earthquake near Zagreb. He noticed two distinct sets of P-wave and S-wave arrivals at epicentral distances greater than ~200 km: a slower direct wave propagating through the upper crust (P_g, S_g) and a faster wave that arrived earlier at distant stations (P_n, S_n).

He correctly deduced that the earlier arrivals traveled along a deeper, higher-velocity layer. The velocity of compressional waves jumped abruptly from ~6.0–6.8 km/s in the crust to >8.0 km/s in the upper mantle. This primary boundary is named the **Mohorovičić discontinuity (Moho)**.

Statement 4: "Transverse waves do not propagate in outer core (fluid state)" — CORRECT

Seismic body waves consist of two fundamental types:

- **Compressional / Primary (P) Waves:** Longitudinal waves that cause alternating compression and dilation. Velocity: $$V_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$ where K is bulk modulus, μ is shear modulus (rigidity), and ρ is density.
- **Shear / Transverse (S) Waves:** Transverse waves that cause particle oscillation perpendicular to wave propagation. Velocity: $$V_s = \sqrt{\frac{\mu}{\rho}}$$

Fluids (liquids and gases) lack shear strength and cannot sustain elastic shear stress; therefore, their shear modulus is zero ($\mu = 0$). As a mathematical and physical consequence, $V_s = 0$. In 1906, Richard Dixon Oldham and later Beno Gutenberg (1914) documented that S-waves do not transmit across the outer core, generating a massive shadow zone for direct S-waves extending beyond an epicentral distance of 103°. This proves conclusively that the outer core is in a fluid/liquid state.

2. Summary Matrix of Earth's Major Discontinuities

Discontinuity Name	Depth (km)	Discoverer & Year	Physical & Compositional Transition
Conrad Discontinuity	15–20 km (discontinuous)	Victor Conrad (1925)	Upper crust (felsic/granitic) to lower crust (mafic/basaltic).
Mohorovičić (Moho)	~30–35 km (continents) ~7–10 km (oceans)	Andrija Mohorovičić (1909)	Crust (felsic/mafic) to Upper Mantle (peridotitic; V_p jumps from ~6.8 to >8.0 km/s).
Repetti Discontinuity	~660–700 km	William Repetti (1930)	Upper Mantle transition zone to Lower Mantle (ringwoodite $ \rightarrow$ bridgmanite + ferropericlase).
Gutenberg Discontinuity (CMB)	**2890–2898 km**	Beno Gutenberg (1914)	Lower Mantle (solid silicate) to Outer Core (molten liquid Fe-Ni; $V_s \rightarrow 0$).
	5145–5150 km	Inge Lehmann (1936)	

Discontinuity Name	Depth (km)	Discoverer & Year	Physical & Compositional Transition
Lehmann Discontinuity (ICB)			Liquid Outer Core to Solid Inner Core (V_p jumps from ~10.3 to 11.2 km/s, S-waves reappear).

3. Authoritative Textbook References

- **Winter (2014)** *Principles of Igneous and Metamorphic Petrology*, 2nd Ed., Pearson, Chapter 1 ("Some Fundamental Concepts"), Section 1.2 ("The Internal Structure of the Earth"), pp. 6–8, Figs. 1-2 & 1-3.
- **White (2013)** *Geochemistry*, Wiley-Blackwell, Chapter 11 ("The Mantle and Core"), pp. 450–475.
- **Van der Pluijm & Marshak (2004)** *Earth Structure*, 2nd Ed., W. W. Norton, Chapter 1 ("A First Look"), pp. 1–15.
- **Dziewonski & Anderson (1981)** "Preliminary Reference Earth Model (PREM)", *Physics of the Earth and Planetary Interiors*, 25(4): 297–356.